LR-FHSS in LEO

Cross Layered Simulation

- Focus: equations, assumptions, and measurable outputs
- Inputs: updated population data + satellite geometry
- Outputs: nodes, demod states, decoded packets, elevation effects, energy

Problem Statement

What Is Estimated at Each Step

Given satellite state at step t, estimate:

1. covered population $P_{\mathrm{cov}}(t)$
2. active nodes $N_{\mathrm{node}}(t)$
3. demodulator pool $N_{\mathrm{demod}}(t)$
4. elevation-conditioned load and decoding behavior
5. busy/idle/sleep demod split and power

End-to-End Mathematical Chain

- Orbit propagation -> footprint geometry
- Footprint + population catalog -> $P_{\mathrm{cov}}(t)$
- $P_{\mathrm{cov}}(t)$ -> nodes and demodulators
- Elevation model -> busy/idle/sleep states
- Demod states -> power and LR-FHSS decode behavior

Formula 1: Orbit Mean Motion

$$n = \sqrt{\frac{\mu}{a^3}}$$

- Symbols: n mean motion, μ Earth gravitational parameter, a semi-major axis.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Formula 2: Mean Anomaly Update

$$M(t) = M_0 + n(t - t_0)$$

- Symbols: $M(t)$ anomaly at time t, M_0 anomaly at reference epoch t_0, n mean motion.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Formula 3: Horizon Central Angle

$$\psi_h = \arccos\left(\frac{R_E}{r_{\mathrm{orb}}}\right)$$

- Symbols: ψ_h horizon central angle, R_E Earth radius, r_{orb} orbital radius.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Formula 4: Geometric Footprint Radius

$$R_{\text{geo}} = R_E \psi_h$$

- Symbols: R_{geo} geometric footprint radius, R_E Earth radius, ψ_h horizon central angle.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Formula 5: Effective Footprint Radius

$$R_{\mathrm{fp}}(t) = \min\left(R_{\mathrm{geo}}(t),\ R_{\mathrm{cfg}}\frac{h(t)}{h_{\mathrm{cfg}}}\right)$$

- Symbols: $R_{\mathrm{fp}}(t)$ effective footprint radius, R_{cfg} configured radius.
- Symbols: $h(t)$ current altitude, h_{cfg} reference altitude.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Formula 6: Covered Population

$$P_{\mathrm{cov}}(t) = \sum_i p_i \, \mathbf{1}\left[d_i(t) \le R_{\mathrm{fp}}(t)\right]$$

- Symbols: $P_{\mathrm{cov}}(t)$ covered population, p_i population at catalog point i.
- Symbols: $d_i(t)$ distance from footprint center to point i, $\mathbf{1}[\cdot]$ indicator.
- Ref: Natural Earth populated places https://naciscdn.org/naturalearth/10m/cultural/ne_10m_populated_places.zip

Formula 7: Node Mapping

$$N_{\mathrm{node}}(t) = \mathrm{round}\left(P_{\mathrm{cov}}(t)\rho_{\mathrm{node}}\right)$$

- Symbols: $N_{\mathrm{node}}(t)$ estimated active nodes, ρ_{node} node/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Formula 8: Demod Mapping

$$N_{\text{demod}}(t) = \text{round}\left(P_{\text{cov}}(t)\rho_{\text{demod}}\right)$$

- Config values: $\rho_{\text{node}} = 10^{-5}$, $\rho_{\text{demod}} = 10^{-2}$.
- Symbols: $N_{\text{demod}}(t)$ estimated demodulator pool, ρ_{demod} demod/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Formula 9: Mean Slant Range per Elevation

$$\bar{d}_e(t) = \frac{1}{N_e(t)} \sum_{u=1}^{N_e(t)} d_{u,e}(t)$$

- Symbols: e elevation bin, $N_e(t)$ users in that bin.
- Symbols: $d_{u,e}(t)$ user-u slant range, $\bar{d}_e(t)$ mean slant range.
- Ref: Slant-range impact concept (Friis distance dependence) https://doi.org/10.1109/JRPROC.1946.234568

Formula 10: Relative Path-Loss Pressure

$$\phi_e(t) = \left(\frac{\bar{d}_e(t)}{d_{\mathrm{ref}}} \right)^2$$

- Symbols: $\phi_e(t)$ path-loss pressure factor, $\bar{d}_e(t)$ mean slant range.
- Symbols: d_{ref} reference range (in code, tied to satellite altitude).
- Ref: Free-space distance-loss relation (Friis)
  https://doi.org/10.1109/JRPROC.1946.234568

Formula 11: Elevation Load Factor

$$f_e(t) = N_e(t) \cdot \alpha_{\text{act}} \cdot \phi_e(t) \cdot 0.01 N_{\text{demod}}(t)$$

- Symbols: $f_e(t)$ load factor, α_{act} activity ratio.
- Symbols: $N_{\text{demod}}(t)$ demod pool size, $N_e(t)$ users at elevation bin e.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Formula 12: Busy Demodulators

$$N_{\text{busy},e}(t) = \min\left(N_{\text{demod}}(t), \lceil f_e(t) \rceil\right)$$

- Symbols: $N_{\text{demod}}(t)$ total demodulators at step t.
- Symbols: $N_{\text{busy},e}$ busy demods at elevation e, $f_e(t)$ load factor.
- Ref: Implementation model in your code `modules/satellite_stepper.py`

Formula 13: Remaining Demodulators

$$N_{\mathrm{rem},e}(t) = N_{\mathrm{demod}}(t) - N_{\mathrm{busy},e}(t)$$

- Symbols: $N_{\mathrm{rem},e}$ demods not busy, N_{demod} total demods.
- Ref: Implementation model in your code `modules/satellite_stepper.py`

Formula 14: Sleep Demodulators

$$N_{\text{sleep},e}(t) = \text{round}\left(\beta_{\text{sleep}} N_{\text{rem},e}(t)\right)$$

- Symbols: $N_{\text{sleep},e}$ sleep demods, β_{sleep} sleep ratio.
- Ref: Implementation model in your code `modules/satellite_stepper.py`

Formula 15: Idle Demodulators

$$N_{\text{idle},e}(t) = N_{\text{rem},e}(t) - N_{\text{sleep},e}(t)$$

- Symbols: $N_{\text{idle},e}$ idle demods, $N_{\text{rem},e}$ remaining demods.
- Ref: Implementation model in your code `modules/satellite_stepper.py`

Formula 16: Power Model

$$P_e(t) = P_0 + N_{\text{idle},e}(t)P_{\text{idle}} + N_{\text{busy},e}(t)P_{\text{busy}}$$

- Symbols: $P_e(t)$ total modeled power for elevation scenario e.
- Symbols: P_0 baseline power, P_{idle} idle demod power, P_{busy} busy demod power.
- Symbols: $N_{\text{idle},e}(t)$ and $N_{\text{busy},e}(t)$ are stepwise demod state counts.
- Current constants: $P_0 = 2$ mW, $P_{\text{idle}} = 9$ mW, $P_{\text{busy}} = 100$ mW.
- Ref: power-state modeling basis https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-software-defined-radio-terminals_ISLPED-2006.pdf

Future Prediction

Predicting Next Time Steps

Goal: predict busy demodulators and energy at future horizon $t + \Delta$.

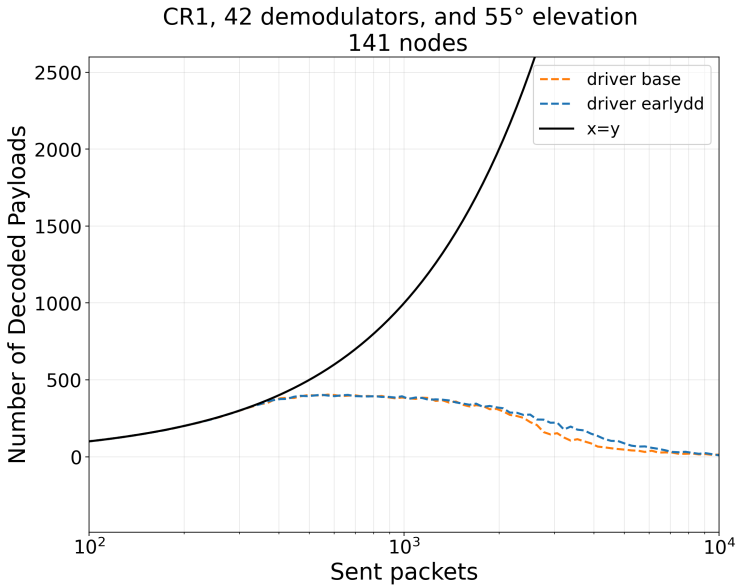
1. Generate future satellite states by stepping orbit index forward.
2. For each future step, recompute footprint and covered population.
3. Map covered population to future `calculated_nodes` and `calculated_demodulators`.
4. For each elevation (90/55/25), estimate future busy/idle/sleep states.
5. Convert those states to future energy using the same power constants.

One-Position Decode Results

Elevation Curves (90 deg)

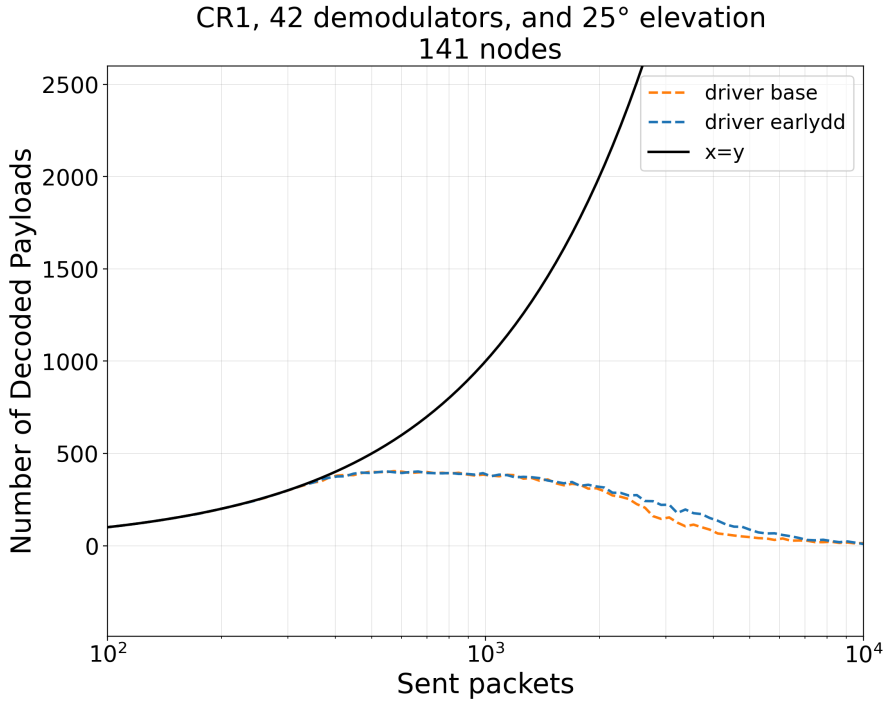
One-Position Decode Results

Elevation Curves (55 deg)



One-Position Decode Results

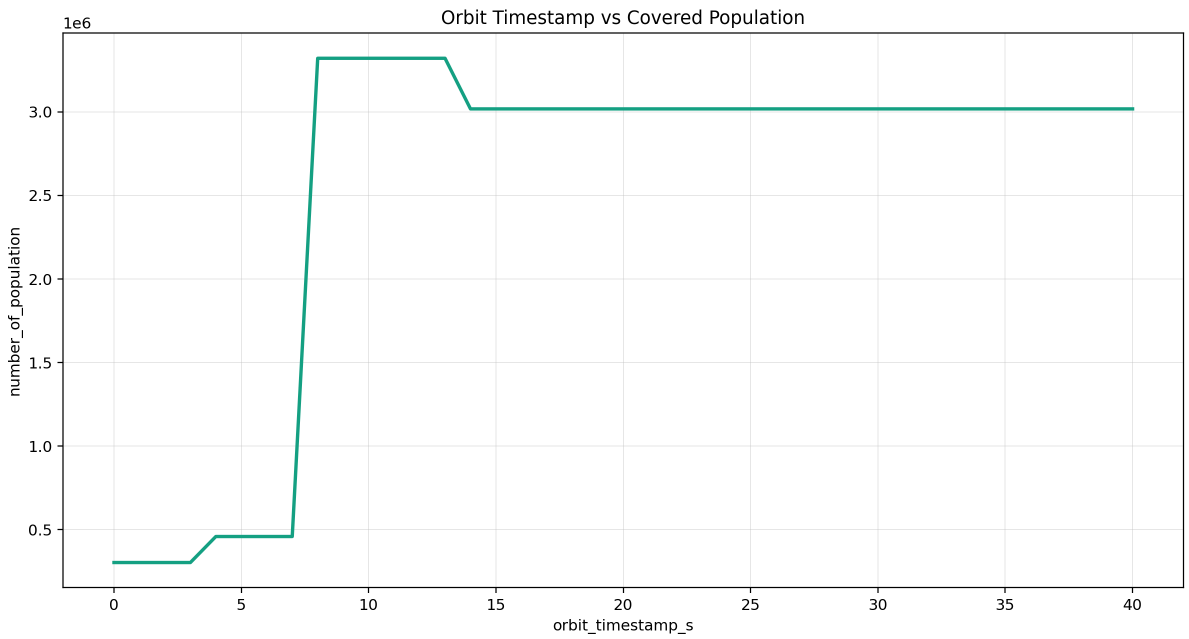
Elevation Curve (25 deg)



- Same demod budget, stronger geometry penalty at lower elevation.

Satellite Stepper Outputs

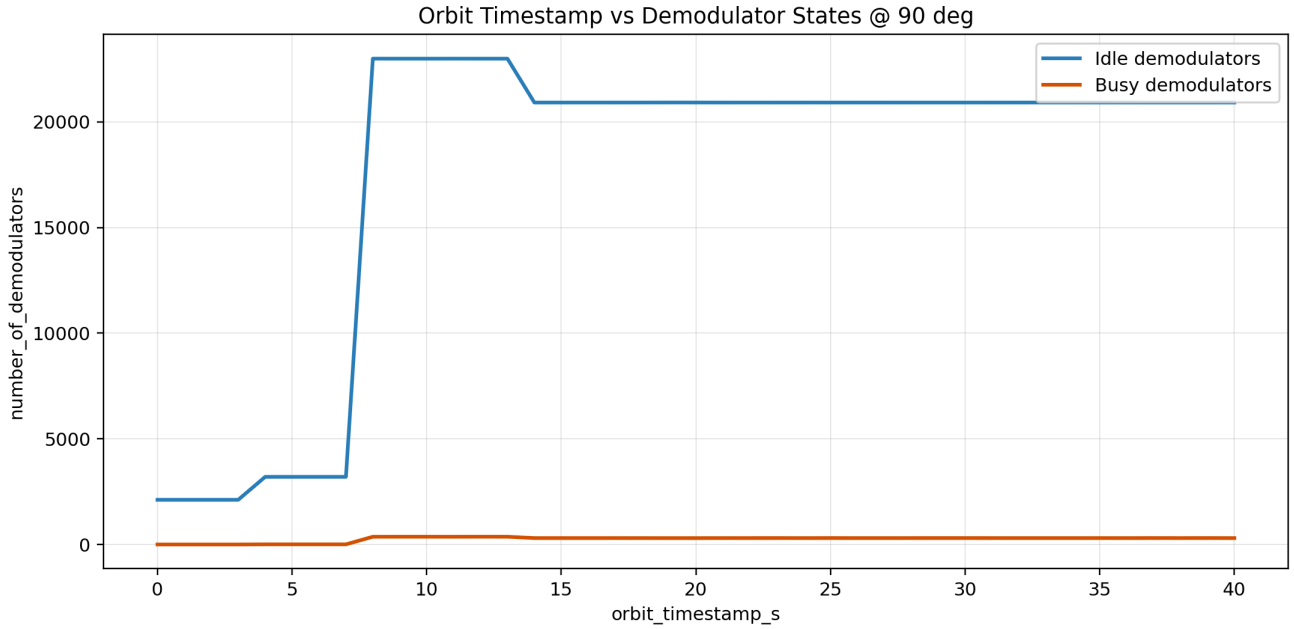
Population and Resource Dynamics



- Per step: covered population, nodes, and demod count.

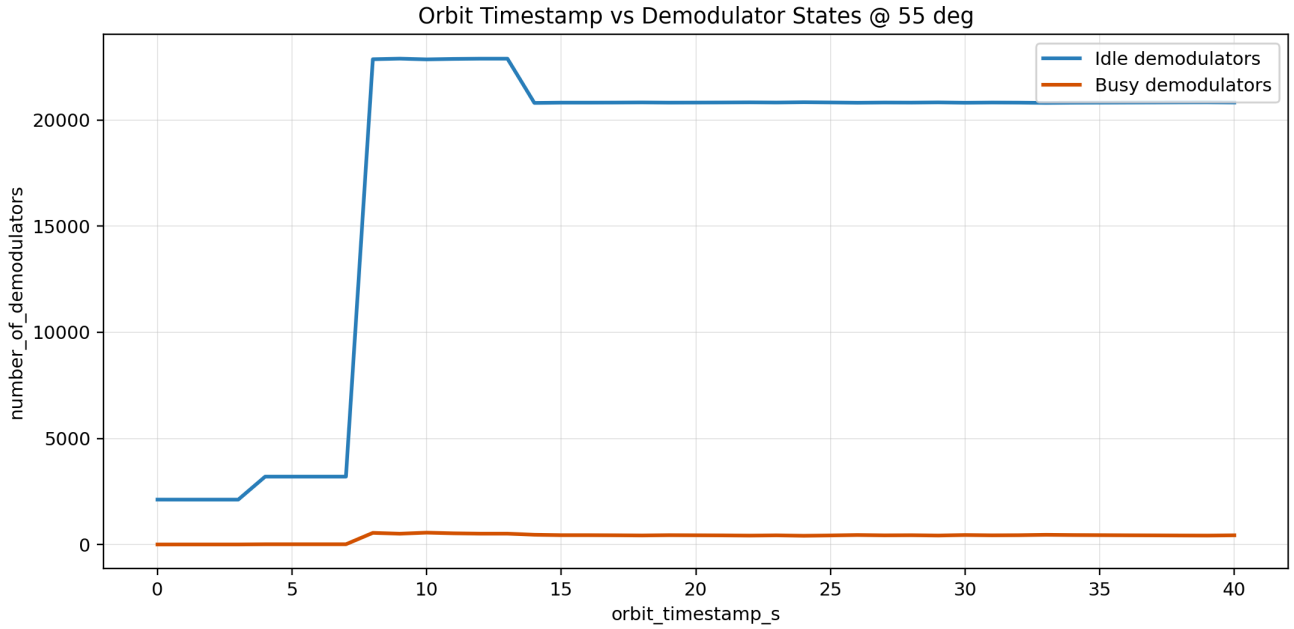
Demodulator State Evidence

Busy/Idle vs Orbit Timestamp (90 deg)



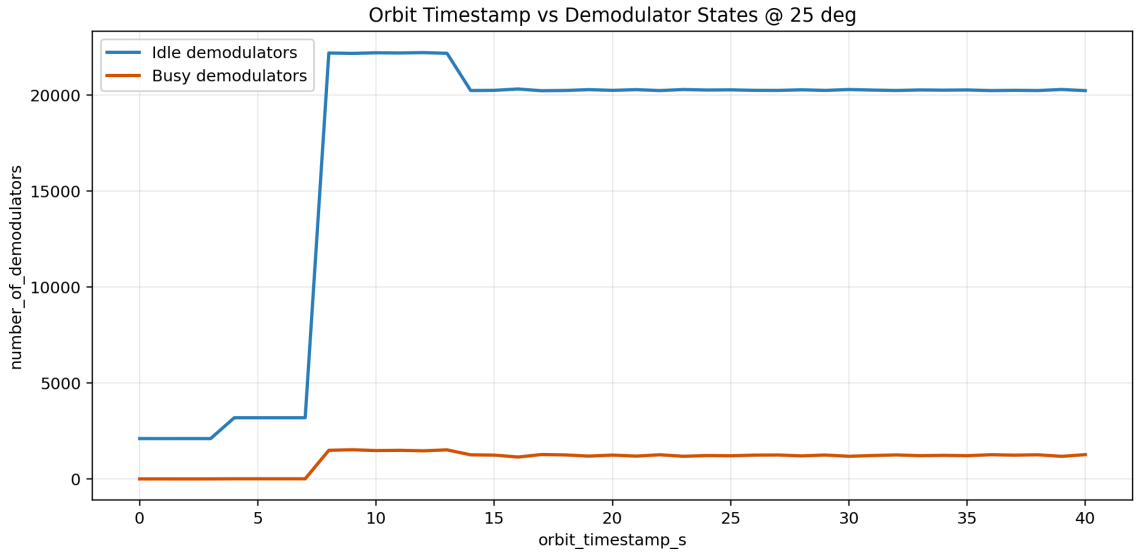
Demodulator State Results

Busy/Idle vs Orbit Timestamp (55 deg)

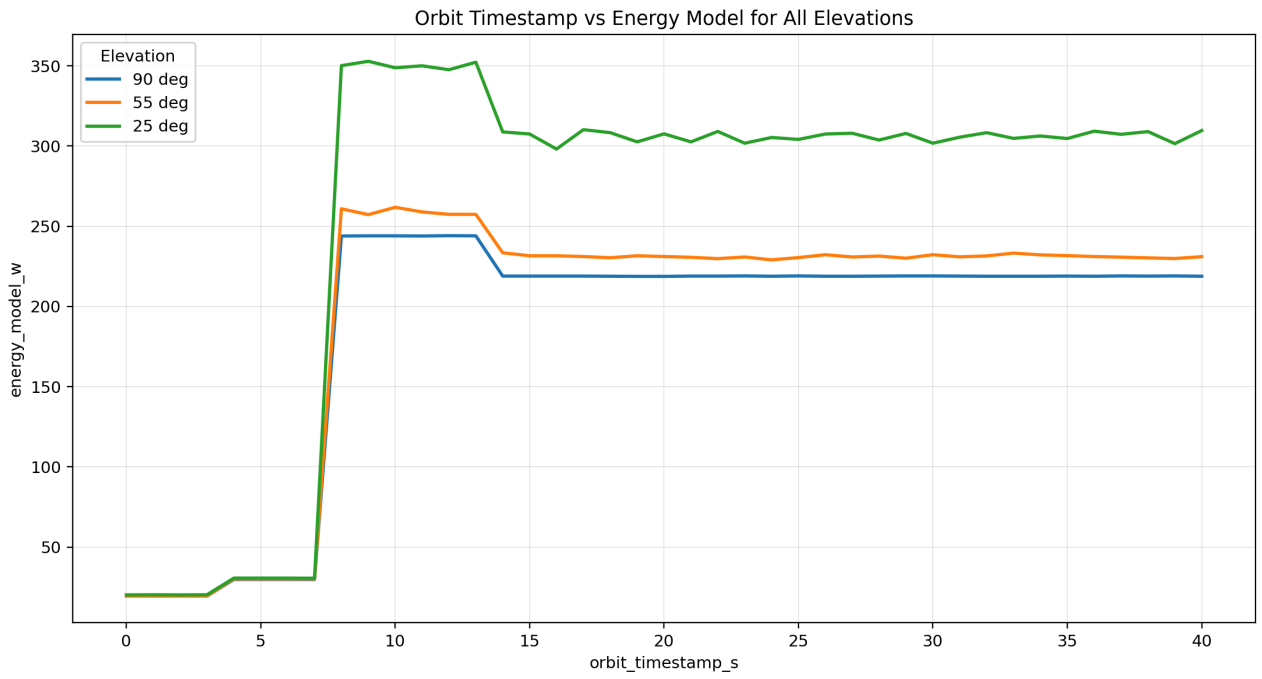


Demodulator State Results

Busy/Idle vs Orbit Timestamp (25 deg)



Energy Model Results (90, 55 and 25 deg)



- Energy follows geometry-driven demod state transitions.

Main Takeaways

1. Geometry controls coverage and distance.
2. Coverage controls node/demod provisioning.
3. Elevation changes distance and busy occupancy.
4. Busy occupancy is the main power driver.
5. One-position decode plots are consistent with this chain.

Citable Paper Sources (URLs)

- LR-FHSS overview paper: https://doi.org/10.1109/MCOM.001.2000627
- 3GPP NTN reference (TR 38.811): https://www.3gpp.org/DynaReport/38.811.htm
- Friis transmission formula: https://doi.org/10.1109/JRPROC.1946.234568
- Shannon communication theory: https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
- ALOHA system paper: https://doi.org/10.1145/1478462.1478502

Citable Data Sources (URLs)

- Natural Earth populated places: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_populated_places.zip
- Natural Earth countries: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_admin_0_countries.zip
- Natural Earth lakes: https://naciscdn.org/naturalearth/10m/physical/ne_10m_lakes.zip
- Natural Earth rivers: https://naciscdn.org/naturalearth/10m/physical/ne_10m_rivers_lake_centerlines.zip
- Demod power baseline reference: https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-software-defined-radio-terminals_ISLPED-2006.pdf